

NURBEK TASTAN

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PROFILE

Doctoral researcher in machine learning with a strong foundation in probabilistic modeling, optimization, and scalable systems. My research focuses on **trustworthy federated learning**, with particular emphasis on **privacy**, **fairness**, **efficiency**, and **robustness** in collaborative settings. I also develop methods to make large-scale models, including **language models**, more efficiently fine-tunable while also preserving utility and user confidentiality.

Passionate about building machine learning systems that are not only performant, but also **privacy-aware**, **equitable**, and **computationally viable** for real-world deployment.

EDUCATION

PhD in Machine Learning, Mohamed bin Zayed University of Artificial Intelligence 2023 - Present
Abu-Dhabi, United Arab Emirates

Relevant Coursework: Foundations and Advanced Topics in Machine Learning, Advanced Probabilistic and Statistical Inference, Advanced Topics in Continuous Optimization, Federated Learning, Safe and Robust Computer Vision, Advanced Machine Learning

MSc in Machine Learning, Mohamed bin Zayed University of Artificial Intelligence 2021 - 2023
Abu-Dhabi, United Arab Emirates

Relevant Coursework: Mathematical Foundations for AI, Machine Learning, Advanced ML, Trustworthy AI, Probabilistic and Statistical Inference, Causality, Reinforcement Learning, Optimization

BSc in Systems of Information Security, International Information Technology University 2017 - 2021
Almaty, Kazakhstan

Relevant Coursework: Programming (Python, C++, Java), ML, Cybersecurity, a lot of Maths
GPA **3.94**/4.00 (Top 1st of 2021 graduating class (out of 900 students))

EXPERIENCE

Lead Teaching Assistant Jan 2022 - Present
MBZUAI *Abu-Dhabi, United Arab Emirates*

- Courses: [DS702: Big Data Processing], [MTH701: Mathematical Foundations of Artificial Intelligence]
- Led sessions for graduate courses, covering distributed data systems and mathematical foundations for AI.
- Designed problem sets and lab materials in probability theory, statistical inference, and scalable ML systems.
- Supervised student projects and delivered guidance on Spark, Hadoop, PageRank, LSH, and big data workflows.
- Enhanced student understanding through hands-on tutorials and assessment-driven feedback.
- More information on <https://tnurbek.github.io/teaching/>.

Data Scientist May 2020 - Sep 2021
Healthcare IT *Almaty, Kazakhstan*

- Developed classification models for highly imbalanced healthcare data.
- Built OCR-based object detection pipelines for extracting clinical information from scanned documents.
- Designed and maintained a big data pipeline using Apache Spark, Airflow, and Hadoop.

Python Instructor Jan 2018 - Mar 2019
Bolashak School, IITU *Almaty, Kazakhstan*

- Taught programming fundamentals and advanced topics (OOP, web scraping, algorithms) in Python and C++.
- Built a full-stack school management system using Python and Django to streamline internal workflows.

SKILLS

Languages	Python, C++, Scala, SQL (PostgreSQL, ClickHouse, Cassandra)
Frameworks	PyTorch, Tensorflow, Spark/PySpark, HuggingFace Transformers
Research Focus	Federated Learning, Decentralized Learning, Trustworthy Machine Learning, Privacy, Fairness, Differential Privacy, Data Valuation, Responsible AI, Large-Scale Optimization, Distributed Optimization, Efficient Machine Learning, Efficient Fine-tuning, LLMs, PEFT, Sparse Models / Sparsity in LLMs, On-Device Learning, Scalable ML, AI Ethics
Additional Tools	Big Data: Apache Spark, PySpark, Hadoop, Kafka, Airflow, Spark Streaming, Tools & Infrastructure: Docker, Git, Unix/Linux, REST API, TensorBoard, W&B

PUBLICATIONS

Stochastic Self-Organization in Multi-Agent Systems

*Nurbek Tastan, Samuel Horváth, Karthik Nandakumar. **ICLR, 2026.** Under Review.*

LoFT: Low-Rank Adaptation That Behaves Like Full Fine-Tuning.

Nurbek Tastan, Stefanos Laskaridis, Martin Takáč, Karthik Nandakumar, Samuel Horváth.

ICML 2025 Workshop, Tiny Titans: TTODLer-FM@ICML. **Honorable Mention.**

***ICLR, 2026.** Under Review.*

MoLM: Mixture of LoRA Markers

*Samar Fares, Nurbek Tastan, Noor Hussein, Karthik Nandakumar. **ICLR, 2026.** Under Review.*

CYCLE: Choosing Your Collaborators Wisely to Enhance Collaborative Fairness in Decentralized Learning.

*Nurbek Tastan, Samuel Horváth, Karthik Nandakumar. **TMLR, 08/2025.***

A Framework for Double-Blind Federated Adaptation of Foundation Models.

*Nurbek Tastan, Karthik Nandakumar. **ICCV, 2025.** ICLR 2025 Workshop (MCDL@ICLR25).*

Aequa: Provably Fair Federated Learning using Slimmable Networks.

*Nurbek Tastan, Samuel Horváth, Karthik Nandakumar. **ICML, 2025.***

FedPeWS: Personalized Warmup via Subnetworks for Enhanced Heterogeneous Federated Learning.

*Nurbek Tastan, Samuel Horváth, Martin Takáč, Karthik Nandakumar. **CPAL, 2025. PMLR.***

Redefining Contributions: Shapley-Driven Federated Learning.

*Nurbek Tastan, Samar Fares, Toluwani Aremu, Samuel Horváth, Karthik Nandakumar. **IJCAI, 2024.***

Collaborative Learning of Anomalies with Privacy (CLAP) for Unsupervised Video Anomaly Detection: A New Baseline.

*Anas Al-lahham, Muhammad Zaigham Zaheer, Nurbek Tastan, Karthik Nandakumar. **CVPR, 2024.***

A Coarse-to-Fine Pseudo-Labeling (C2FPL) Framework for Unsupervised Video Anomaly Detection.

*Anas Al-lahham, Nurbek Tastan, Muhammad Zaigham Zaheer, Karthik Nandakumar. **WACV, 2024.***

CaPriDe Learning: Confidential and Private Decentralized Learning based on Encryption-friendly Distillation Loss.

*Nurbek Tastan, Karthik Nandakumar. **CVPR, 2023.***

HONORS & AWARDS

National & Prestigious Awards

Presidential Scholarship, Ministry of Education, Kazakhstan	2020
Award “Altyn Belgi” (Golden Badge), Ministry of Education, Kazakhstan	2017
National Presidential Olympiad, Ministry of Education, Kazakhstan (I place - regional, III place - final)	2016
Republican Scientific Competition (Applied Mathematics) - III place (Bronze)	2016
National Scientific Competition (Computer Science) - II place (Silver)	2020, 2021
National Mathematics competition among students, Kazakhstan, IV place	2019

International, Technical Competitions & Awards

ICML 2025 TTODLer-FM Workshop, Honorable Mention	2025
ICPC regional, II place	2019
“Hack The Space” Hackathon, Dubai, United Arab Emirates, “The Best Math Team”	2022

University- or Region-level Competitions

Scientific competition (Computer Science), IITU, Kazakhstan, II place	2021
Scientific competition (Computer Science), IITU, Kazakhstan, III place	2020
Mathematics competition, IITU, I place	2019
Mathematics competition, IITU, II place	2018
Physics Olympiad, II place (Silver)	2015
Regional scientific competition (Applied Mathematics), I place	2015
Mathematics Olympiad, I place (Gold)	2013

CONFERENCE / WORKSHOP PRESENTATIONS

– Aequa: Provably Fair Federated Learning using Slimmable Networks.	ICML , Vancouver, Canada, Jul 2025.
– LoFT: Low-Rank Adaptation That Behaves Like Full Fine-Tuning.	ICML TTODLer-FM Workshop , Vancouver, Canada, Jul 2025.
– A Framework for Double-Blind Federated Adaptation of Foundation Models.	ICLR MCDC Workshop , Singapore, Apr 2025.
– FedPeWS: Personalized Warmup via Subnetworks for Enhanced Heterogeneous Federated Learning.	CPAL , Stanford, USA, Mar 2025.
– Confidential, Private, and Fair Decentralized Learning.	ADIA Symposium , Abu Dhabi, UAE, Nov 2024.
– Redefining Contributions: Shapley-Driven Federated Learning.	IJCAI , Jeju, South Korea, Aug 2024.
– Collaborative Learning of Anomalies with Privacy (CLAP) for Unsupervised Video Anomaly Detection: A New Baseline.	CVPR , Seattle WA, USA, Jun 2024.
– A Coarse-to-Fine Pseudo-Labeling (C2FPL) Framework for Unsupervised Video Anomaly Detection.	WACV , Waikoloa, Hawaii, USA, Jan 2024.
– CaPriDe Learning: Confidential and Private Decentralized Learning based on Encryption-friendly Distillation Loss.	2nd MBZUAI CL Workshop , Abu Dhabi, UAE, Dec 2023.
	CVPR , Vancouver, Canada, Jun 2023.

ACADEMIC SERVICES

Program Committee (PC) / Reviewer for:

Conference: AAAI 2024, ICLR 2025, CVPR 2025, ICML 2025, ICCV 2025, NeurIPS 2025,
WACV 2026, AAAI 2026, ICLR 2026

Journal: IEEE Transactions on Mobile Computing (TMC), IEEE Signal Processing Letters (SPL),
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)

CERTIFICATES

Machine Learning Specialization by DeepLearning.AI	Coursera, Dec, 2022
– Supervised Machine Learning: Regression and Classification	
– Advanced Learning Algorithms	
– Unsupervised Learning, Recommenders, Reinforcement Learning	
Cybersecurity by IBM	Coursera, Jun, 2020

REFERENCES

Dr. Karthik Nandakumar
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Associate Professor at MBZUAI & MSU
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Dr. Samuel Horváth
PhD Supervisor
Assistant Professor at MBZUAI
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